

agentic-ai-azure-semantic-kernel

Companion - Semantic Kernel on Azure

Agentic AI on Azure - Enterprise Master Class | Handout (slides + notes)

Notes

Welcome. This deck covers 'agentic-ai-azure-semantic-kernel' - Companion - Semantic Kernel on Azure. Frame the problem and why it matters, then walk the class through the learning goal, the enterprise use case, the architecture/flow, the key concepts, a live run in VS Code, and the architect's takeaways. The repo runs locally with no Azure, so demo it live.

Slide 2 - Learning goal

- Compose AI skills with a Kernel + plugins
- Expose a native `@kernel_function`
- Compare Semantic Kernel to the other frameworks

Notes

[Learning goal] Walk through these talking points:

- Compose AI skills with a Kernel + plugins
- Expose a native `@kernel_function`
- Compare Semantic Kernel to the other frameworks

Ask the class: which of these ideas is new to you?

Slide 3 - Enterprise use case

- Claims Validation with a native plugin
- PolicyPlugin validates a policy number
- Same 'tool the agent calls' idea, in SK

Notes

[Enterprise use case] Walk through these talking points:

- Claims Validation with a native plugin
- PolicyPlugin validates a policy number
- Same 'tool the agent calls' idea, in SK

Tip: relate this to a regulated scenario your students know.

Slide 4 - Architecture / flow

- Kernel + `add_plugin(PolicyPlugin, 'policy')`
- Mock: `kernel.invoke(plugin, function)`
- Azure: `AzureChatCompletion` + auto function calling
- Structured fields stay authoritative

Notes

[Architecture / flow] Walk through these talking points:

- Kernel + `add_plugin(PolicyPlugin, 'policy')`
- Mock: `kernel.invoke(plugin, function)`
- Azure: `AzureChatCompletion` + auto function calling
- Structured fields stay authoritative

Demo: trace a single request through these steps live.

Slide 5 - Key concepts

- @kernel_function plugins; KernelArguments
- Runs for real offline via kernel.invoke
- SK is converging into Agent Framework 1.0

Notes

[Key concepts] Walk through these talking points:

- @kernel_function plugins; KernelArguments
- Runs for real offline via kernel.invoke
- SK is converging into Agent Framework 1.0

Open the named file in the repo while you explain each point.

Slide 6 - Run it (VS Code - no Azure needed)

- git clone the repo, then open it in VS Code
- python -m venv .venv -> activate it
- pip install -r requirements.txt
- uvicorn app.main:app --reload
- Open /docs and call POST /api/v1/claims/validate
- Runs in MOCK mode - no Azure/keys needed

Notes

[Run it (VS Code - no Azure needed)] Walk through these talking points:

- git clone the repo, then open it in VS Code
- python -m venv .venv -> activate it
- pip install -r requirements.txt
- uvicorn app.main:app --reload
- Open /docs and call POST /api/v1/claims/validate
- Runs in MOCK mode - no Azure/keys needed

Pause for questions before moving on.

Slide 7 - Architect's takeaways

- Great for embedding skills into existing apps
- Strong across .NET and Python
- Skills carry into the Agent Framework labs

Notes

[Architect's takeaways] Walk through these talking points:

- Great for embedding skills into existing apps
- Strong across .NET and Python
- Skills carry into the Agent Framework labs

Discussion: when would you NOT choose this approach?